

# Rajat Saxena

✉ [Rajat.Saxena@campus.lmu.de](mailto:Rajat.Saxena@campus.lmu.de) | 🌐 Portfolio

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## ABOUT

Master's student at LMU Munich with experience developing simulations in theoretical physics and astrophysics. My research interests lie in quantum many-body physics, statistical mechanics and the physics of machine learning. I am also passionate about science outreach, philosophy and history.

## EDUCATION

- **MSc in Physics** October 2024 - Present  
*Ludwig Maximilian University (LMU) of Munich; Grade: 2.24 (1 Highest, 5 Fail)*  
*Advisor: Dr. Steffen Rulands, Professor, LMU*  
Munich, Germany
- **BSc Blended Physics** June 2024  
*Savitribai Phule Pune University, CGPA: 9.32 (10 Highest, 4 Fail)*  
*Advisor: Dr Deepak Dhar, Distinguished Professor Emeritus, IISER - Pune*  
Pune, India

## RESEARCH EXPERIENCE

- **Arnold Sommerfeld Centre for Theoretical Physics, LMU Munich** October 2025 - Present  
*Master Thesis*  
Mentor: Prof. Dr Steffen Rulands, Onurcan Bektas
  - Investigating the emergence of self-organised hierarchies in robotic systems governed by deep neural networks (DNNs) for decision-making.
  - Assembling and programming autonomous robots to perform collective tasks. Implementing multiple neural network architectures trained on multimodal sensory inputs.
  - Designing and simulating experiments to model and analyse collective behaviour and performance outcomes under different tasks.
- **Universitäts-Sternwarte München (University Observatory Munich)** March 2025 - Present  
*Research Intern*  
Mentor: Dr Tilman Birnstiel, Professor
  - Developed N-body simulations to model the gravitational collapse of a dust cloud in the presence of a central star and a pre-formed gas disk.
  - Simulated particle trajectories integrated with gas dynamics during protostar formation, leveraging hydrodynamic snapshots from (from [Tārā](#)) to interpolate density, velocity, and temperature fields in both time and space at particle locations.
  - Implemented semi-implicit and exponential midpoint integration schemes in Python and performed numerical simulations on a self-built codebase, achieving faster execution than REBOUND.
  - Summarised the group's research for Astrobites, translating complex results into accessible articles. Available: [\[🌐\]](#)
- **Indian Institute of Science Education and Research (IISER) – Pune** August 2023 - May 2024  
*Bachelor Thesis*  
Mentor: Dr Deepak Dhar, Distinguished Professor Emeritus
  - Studied foundational statistical mechanics and phase transitions through standard texts to build a strong theoretical base.
  - Developed a Python simulation of a 1D zero-temperature random field Ising model to investigate discrete magnetisation jumps in ferromagnetic materials with impurities and defects.
  - Analysed the impact of lattice size, random field standard deviation, and probability distributions on magnetisation curves, quantifying parameter-dependent behaviour.
- **Inter-University Centre for Astronomy and Astrophysics (IUCAA)** June 2023 - July 2023  
*Summer Intern*  
Mentor: Mr Ashish Mhaske, Scientific and Technical Officer
  - Led a student team to measure 21-cm neutral hydrogen power spectrum across different Galactic longitudes using a conical horn antenna.
  - By analysing Doppler-shifted HI line profiles, we derived the Milky Way's rotation curve and observed the discrepancy between the theoretically predicted and the observed rotation curves.

- Designed and implemented a Python data-analysis pipeline to process source and calibration data.

## • **National Centre for Radio Astrophysics (NCRA) - TIFR**

September 2022 – April 2023

Research Intern

Mentor: Dr Yashwant Gupta, Distinguished Professor and Centre Director

- Developed Python code for folding and binning time-series data to generate pulsar profiles for the Crab Pulsar using uGMRT data.
- Implemented C routines to define on/off gates, enabling precise separation of pulsar on-pulse and off-pulse emission intervals.
- Applied pulsar timing, dedispersion techniques, and correlator analysis, leveraging software tools such as TEMPO2 and GMRT Pulsar Tool (gptool).

## AI/ML PROJECTS

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### ◦ **CNNs for Mass Estimation of X-Rays Clusters**

Employed Convolutional Neural Networks (CNNs) to estimate Galaxy cluster masses from simulated X-ray images and redshift data. The redshift values were concatenated to the convoluted and flattened image data before being fed to the fully connected network.

### ◦ **Emergent Weight Morphologies in Deep Neural Networks**

Investigated the emergent structural patterns in neural network weights that arise independently of the training data. Also examined how these structural patterns change with alterations in hyperparameters and training strategies.

### ◦ **Neural Networks for Quantum Many-Body Physics**

Implemented a Restricted Boltzmann Machine (RBM) ansatz within a Monte Carlo framework to approximate the ground state of a Rydberg atom chain. Utilised the expressive capacity of RBMs to represent many-body wavefunctions and employed stochastic gradient optimisation to minimise the variational energy.

### ◦ **Simulation-Based Inference for Bayesian Cosmological Data Analysis**

Developed a simulation-based inference framework using autoencoder compression and normalising flows to recover cosmological parameters from Gaussian random fields. Benchmarked learned posteriors against analytic Fourier-space likelihoods, identifying effects of compression loss and autoencoder overfitting.

## OUTREACH EXPERIENCE

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### • **LIGO – India Education and Public Outreach (LI-EPO), IUCAA**

January 2023 – July 2024

Intern

Mentor: Dr Debarati Chatterjee, Associate Professor and Chair LI-EPO

- Developed outreach content and social media posts to communicate LIGO's science, and maintained the interferometer demonstrations for public engagement.
- Conducted awareness campaigns across Indian colleges, promoting the upcoming observatory.
- Designed and led an introductory Python workshop for undergraduates, and assisted in organising Astronomy Teacher Training Workshops and local outreach at Hingoli (LIGO India site).

### • **Science Popularisation Centre (SciPop), IUCAA**

Volunteer

- Operated telescopes and led stargazing sessions for school students and the general public, and conducted training for new volunteers in telescope operation.

## MENTORSHIP EXPERIENCE

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- **Teaching Assistant at Quest Classes, Pune** (Aug–Oct 2024): Taught 12th-grade Physics (optics) and 11th-grade Mathematics (inequalities, combinatorics, binomial theorem).

- **Co-founder & Research Head, IDSS Space Society** (Oct 2021 – Jul 2023): Organised guest lectures, stargazing nights, school outreach programs, and space-themed MUNs. Recognised as a "Registered Space Tutor of ISRO" by CBPO, ISRO.

## SELECTED CONFERENCES

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Oral Presentations

- **30th Young Scientists' Conference on Astronomy and Space Physics**, National University of Kyiv April 2024
- **10th National Student Symposium on Physics**, Indian Association Physics Teachers (IAPT) October 2023

Poster Presentations

- **Symposium on Magnetism and Spintronics**, Indian Institute of Technology - Bombay July 2024
- **National Space Science Symposium**, Indian Space Research Organization (ISRO) February 2024

## RELEVANT WORKSHOPS

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- **German Italian Physics Exchange**, German Physical Society & Associazione Italiana Studenti di Fisica *Sept 2025*
- **Summer University for Plasma Physics and Fusion Research**, Max Planck Institute for Plasma Physics *Sept 2025*
- **Summer School in Theoretical (Astro)Physics**, St Xavier's College & IUCAA *June 2024*
- **Course on Nuclear Experiments**, HBCSE-TIFR *March 2024*
- **Workshop on Data Science in Astronomy**, IUCAA *December 2023*

## SKILLS

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- **Programming Languages:** Python, Mathematica, Latex
- **Libraries:** Numpy, Scipy, Astropy, REBOUND, Matplotlib, Pandas
- **ML Libraries:** Keras/TensorFlow, Sklearn
- **Telescope Handling:** Dobsonian & Newtonian telescopes (EQ/Altazimuth mounts)
- **Softwares:** LAMMPS, ORCA, Astrometrica
- **Languages:** English (IELTS: 7.5), Hindi (Native), Marathi (Professional)

## AWARDS AND ACHIEVEMENTS

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- **Exceptional Accomplishments in Physics and Science Communication** *March 2024*  
*Interdisciplinary School of Science, Savitribai Phule Pune University*
  - Conferred by the Hon. Vice Chancellor in recognition of outstanding academic performance and science communication initiatives.
  - Acknowledged as a significant departmental honor highlighting excellence in both research and outreach.
- **Telescope Building Grant (INR 60,000)** *2022–23*  
*Savitribai Phule Pune University*
  - Received university funding to design and construct a Newtonian telescope.

## PROFESSIONAL MEMBERSHIPS

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- **Student Member of "German Physical Society"** *2025 - Present*
- **Student Member of "Indian Physics Association"** *2024 - Present*

## PUBLICATIONS

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- [1] **Saxena, Rajat.** (2024). Introduction to Statistical Mechanics and Simulation of Random Field Ising Model. 10.13140/RG.2.2.34379.81443. Available: [🌐]
- [2] **Saxena, Rajat & Padalkar, Gauri & Patil, Yash & Sherkar, Shreenath & Satam, Harsh & Shah, Manasvi & Rajopadhye, Chinmayee.** (2024). Galaxy Rotation Curve Measurements using a Horn Antenna. Available: [🌐]